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Chained to Technology
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# Chained to Technology

Notable technologies such as artificial intelligence (AI) and machine learning (ML) are thriving today. AI and ML are used to increase the efficiency of tasks, provide solutions to problems, and automate tasks. Pattern recognition in these technologies has already made its way into our daily lives. Facial recognition software in our phones such as Face ID utilizes a neural network and deep learning to recognize unique facial features of a person to make unlocking a phone more efficient. Our phones are pocket-sized computers that make it easy to stay connected and online. They cure our boredom, provide us with instant information, and easy access has led to increased prevalence throughout our daily lives. A study conducted in Q3 2021 shows that on average, people spend 3 hours and 15 minutes on their phones per day, displaying an increasing dependence and growing addiction to electric technologies (Howarth, 2022).

## The Internet and Cloud Computing

The Internet was created on January 1, 1983. Originally invented for military purposes, it shortly expanded and quickly became the primary form of communication throughout the entire world. The overwhelming convenience and effectiveness of it also led to the increasing need for computers in the 1960s. The various uses of the internet increased as technology evolved. It has become the primary form of entertainment through ways such as video streaming, browsing, and social media. It’s also made its way into the education system and

workplace with cloud computing. The term cloud computing was first used in a modern context in 2006. It is the concept of on-demand availability of computer system resources. The abstraction of computing, storage, and network infrastructure assembled as a platform allows applications and systems to be used on demand with no hardware requirements from the user (Knorr, 2022). The cloud computing architecture (see figure 1) displays this level of abstraction. Popular cloud computing technologies include Google Drive, Google Docs, and Dropbox. Google Drive and Dropbox allow for the storage of personal files on the cloud. Google Docs allows for real-time collaborative writing utilizing cloud resources. Real-time collaboration is a staple concept necessary for today's academic and professional workflow.

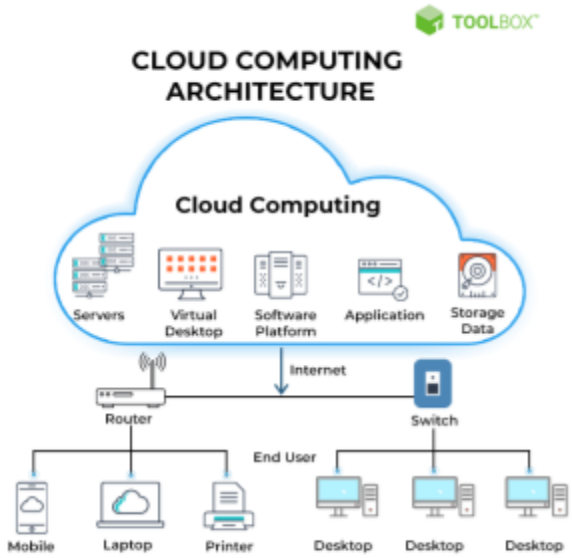


*Figure 1.* Cloud Computing Architecture. https://www.spiceworks.com/tech/cloud/articles/what-is-cloud-computing/

**Types of Cloud Services**

As a service, cloud computing takes form under four different broad categories: infrastructure as a service (IaaS), platform as a service (PaaS), serverless, and software as a service (SaaS) (Microsoft, n.d.). Infrastructure as a service is the most basic category of cloud computing services in which you rent virtual machines from a provider. Platform as a service refers to services that supply on-demand environments for developing software. Serverless computing

overlaps with PaaS, focusing on building app functionality without having to spend time managing servers and infrastructure required to do so. Software as a service is more typically encountered in our daily lives. Cloud providers host the application and the user simply connects to it over the internet. A common form of SaaS is the aforementioned Google Docs.

**Risks of Cloud Dependency**

Without a doubt, cloud computing technology played a major role in maintaining our jobs and helped us survive the armageddon that was the COVID-19 pandemic. Had not been for cloud computing, our reliance on computer and electrical technologies would have made workflow during the pandemic immeasurably more challenging. Unfortunately, businesses that couldn't function remotely saw closure. From that perspective, our dependency on technology is proven to be excessive. Our dependency threatens us with the potential loss of access to cloud services and the loss of data stored on the cloud. A notable example of this occurred in 2017 when Amazon Web Services (AWS) famously went down (Jones, 2020). Naturally, websites and services that relied on it like Grammarly, Medium, Slack, and Trello went down with it. This is a striking indication of how businesses of all sizes are affected by the reliance on technology and their third-party providers (Jones, 2020). In regards to data loss, Microsoft for example does not accept liability in the uneventful loss of your data. In their agreement, it is explicitly stated that "...all online services suffer occasional disruptions and outages, and Microsoft is not liable for any disruption or loss you may suffer as a result (Microsoft, 2022). The widespread adoption of cloud computing has accelerated job productivity but also lead to more occurrences of events similar to the outage that befell AWS.

**Internet for Entertainment**

Today technology plays a major role in our daily lives besides just the workplace. It provides us with information, and entertainment, helps us carry out tasks, and more. A study conducted at Penn State (see figure 2) indicated 77% of people believe society as a whole relied too much on society to succeed (Jonathan et al., 2015). The convenience of technology paves the way to depend on it. Instant communication through the internet allows anyone to quickly find the answer to any question they have throughout the day. Technology is constantly improving and evolving and as convenience grows, it soon begins taking over the time in our day. GSMA reports that mobile device usage will continue to increase for the next three years (The Mobile Economy, 2022). It was only in recent years that the concept of passive entertainment was created. Simply put, passive entertainment means you are on autopilot (Harper, 2020). Watching Netflix or tv, scrolling through social media, or surfing the internet are all various forms of passive entertainment. Free time has been decreasing and people tend to find themselves in "time poverty" (Giurge et al., 2020). The convenience of passive entertainment then fills the void. In 2020, on average, Netflix recorded its subscribers watch 3.2 hours of content (see figure

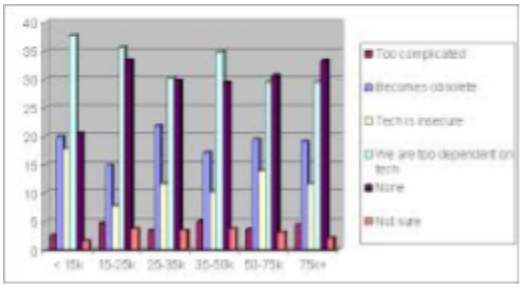


*Figure 2.* Penn State study depicting reliance of technology. https://sites.psu.edu/siowfa15/2015/10/23/is-society-too-dependent-on-computersphones/

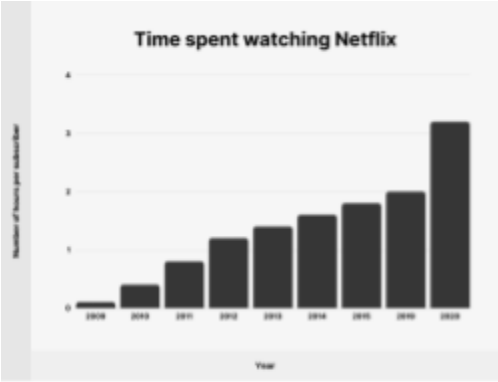


*Figure 3.* Time spent watching Netflix. https://backlinko.com/netflix-users

3) daily (Dean, 2021). This number has steadily grown since 2009 where the users previously watched 0.1 hours daily (Dean, 2021).

**Artificial Intelligence and Machine Learning**

Our reliance on technology also bleeds into the concepts of artificial intelligence and machine learning. At its core machine learning is pattern recognition where an AI agent learns through repetition. Machine learning lies underneath what we see throughout our entire day. It finds and recommends us videos and shows based on what we've watched already, cleans our email inboxes by identifying spam, recommends products based on previous purchases, and more. It is a large convenience in making our lives more efficient, especially when time is so scarce. Machine learning also improves our safety, preventing fraudulent transactions by tracking locations of previous purchases and declining or requesting authorizations when you visit someplace new. Additionally, self-driving or autonomous cars are becoming increasingly popular. A study conducted in 2021 predicted the self-driving car market to grow (see figure 4) from 20.3 million units to 62.4 by 2030 (Markets and Markets, n.d)

The primary factor in its explosive growth is the demand for safer driving experiences and driving assistance. Assistive driving technologies

*Figure 4.* Automotive Vehicle Market Growth. https://www.marketsandmarkets.com/Market-Reports/near-autonomous-passenger-car-market-1220.html

for today's self-autonomous vehicles have proven to be effective. Automatic emergency braking (AEB) is present in most cars produced in 2020 (see figure 5) because of a guideline adopted by auto manufacturers in 2016 (Loeb, 2020). A study showed promising results where AEB prevented 83% of rear-end crashes, a large increase compared to previous reports of 38% to 70% before AEB (Loeb 2020). AEB prevented 57% of rear-end crashes between teenagers, 81% of young adult rear-ended crashes, 100% of adult rear-ended crashes and 71% of older adult rear-ended crashes (Loeb, 2020). However, AEB is acknowledged to be less effective in preventing rear-end crashes occurring in poor weather conditions or at higher speeds (Loeb, 2020). We have proven to be safer in the hands of assisted driving compared to having full manual control over automotive vehicles.

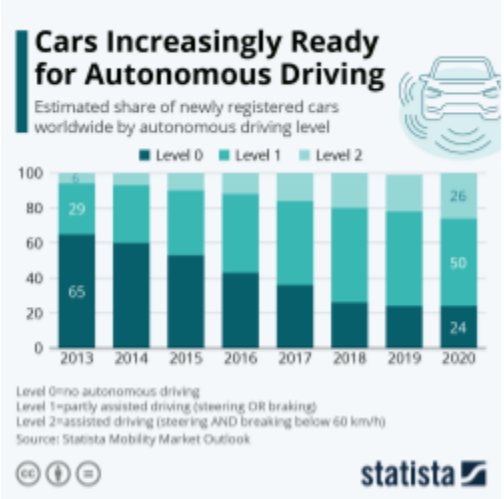


*Figure 5.* Cars Increasingly Ready for Autonomous Driving. https://www.statista.com/chart/25754/newly-registered-cars-by-autonomous-driving-level/

**Presence of Technology**

People depend on technology and thus society has built around that pitfall. It solves and creates many issues. The foundation of work today relies heavily on technology and the internet services it provides. Specialized jobs are created by technology while jobs requiring simple labor can be automated with a robot. Our primary form of entertainment is through technology

with streaming alone taking up 3 hours of our day. Even the roads would be safer when not in the hands of our own. Technology underlies our safety, entertainment, work productivity, and education, and we're chained to it.

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Markets and Markets "Automotive Vehicle Market Growth" infographic showcases the globally increasing demand for autonomous vehicles. It details the trends and demands of the market and how these principles promote market growth. Markets and Markets project the self-driving car market size to grow from 20.3 million units in 2021 to 62.4 million units by 2030.

"Bar Chart." 23 October 2015, Pennsylvania State University, https://sites.psu.edu/siowfa15/2015/10/23/is-society-too-dependent-on-computersphones/

David Jonathan's bar chart showcases the results of a study conducted about society's general opinion about computer and technological dependency. Overall, it is shown that most individuals agree that we are too dependent on tech.

Buchholz, Katharina. "Cars Increasingly Ready for Autonomous Driving." 13 September 2021, Statista. https://www.statista.com/chart/25754/newly-registered-cars-by-autonomous-driving-level/

Katharina Buchholz's column chart showcases the rate of change in vehicles becoming increasingly ready for autonomous driving. The chart shows the decreasing market share trend for Level 0 (no autonomous driving) cars between 2013 to 2020. Level 0 cars fell from taking 65% market share in 2013 to 24% in 2020.

Patil, Prajakta & BasuMallick, Chiradeep. "Cloud Computing Architecture." 9 February 2022, SpiceWorks,

www.spiceworks.com/tech/cloud/articles/what-is-cloud-computing/

Prajakta's and Chiradeep's cloud computing architecture diagram shows the levels of abstraction within cloud computing. The diagram displays that the cloud hosts services such as data storage, applications, software, and servers. End-users can access these services through the internet using networking devices like routers or switches.

Dean, B. "Time Spent Watching Netflix." 7 October 2021, Backlinko. https://backlinko.com/netflix-users

Dean's bar chart tracks the history of Netflix users' growing consumption of content from 2009 until 2020. The chart displays the rapid growth of the consumer's average watch time from 0.1 hours per day in 2009 to 3.2 hours per day in 2020.

Dean, B. (2021, October 7). *Netflix subscriber and Growth Statistics: How Many People Watch Netflix in 2022?* Backlinko. Retrieved November 8, 2022, from https://backlinko.com/netflix-users

Through extensive research, Dean has compiled various graphs showcasing Netflix's rapid growth in popularity between 2009 to 2020. Dean has analyzed metrics such as concurrent subscribers, the influx of new subscribers, subscribers sorted by region, cost of production, and average watch time.

Giurge, L. M., Whillans, A. V., & West, C. (2020). Why Time Poverty Matters for individuals, organizations and nations. *Nature Human Behaviour*, *4*(10), 993–1003. https://doi.org/10.1038/s41562-020-0920-z

Giurge, Whillans, and West research growing societal pains about the conceptual "time poverty", the feeling of having too many things to do and not enough time to do them. They

dive into systemic issues enforced by policymakers and large organizations whose repercussions are felt by the working class. From the perspective of psychologists, Giurge, Whillans, and West analyze and find solutions to the ongoing issue of time poverty.

Harper, H. (2020, September 28). *How "passive entertainment" is only a recent concept, historically speaking*. The Career Project. Retrieved November 8, 2022, from https://www.thecareerproject.org/blog/passive-entertainment-as-a-recent-concept/

Harper defines the recent concept of “passive” entertainment and draws comparisons from it to “active” entertainment. Harper argues that a more engaging, active form of entertainment is where time is better spent when it comes to free time. Generally, she concludes that regardless of whether active or passive, it is all-around important for personal health to relax once in a while.

Howarth, J. (2022, September 16). *Time spent using smartphones (2022 statistics)*. Exploding Topics. Retrieved November 8, 2022, from https://explodingtopics.com/blog/smartphone-usage-stats

Howarth compiles his extensive research surrounding the globally increasing rate of smartphone usage. Haworth displays these statistics using various charts, diagrams, and tables. The measured statistics include, time spend using smartphones in 2022, time spent by region, and smartphone use in the United States. Haworth concludes that screen time is evidently on the rise and whether it is positive or negative is debatable.

Jonathan, D, et al. (2015, October 23). *SIOWFA15: Science in our world: Certainty and controversy*. SiOWfa15 Science in Our World Certainty and Controversy. Retrieved

November 8, 2022, from https://sites.psu.edu/siowfa15/2015/10/23/is-society-too-dependent-on-computersphones/

Jonathan et al. conducted a study where people were asked to give a personal opinion on whether they believe society has become too dependent on (electrical) technology. The results of the study were overwhelmingly one-sided where 77% of participants agreed that we have grown to be too dependent on tech. They highlight the negative impacts of how jobs and society have built around these technologies and that "the importance of enjoying yourself without technology has been lost."

Jones, M. (2020, September 14). *What's the risk of becoming too cloud-dependent?* TechHQ. Retrieved November 8, 2022, from https://techhq.com/2020/09/whats-the-risk-of-becoming-too-cloud-dependent/

Cloud computing technology is an important staple in today's workflow. Jones highlights the issues and dangers of becoming too dependent on these technologies. The COVID-19 pandemic would've been significantly more challenging if not for remote access to work. Jones explains the large danger lies in relying on cloud service providers. He highlights instances where Google and Amazon's own cloud services went down and took down other key services with them. Loss of access to services like the Google Drive suite and Amazon Web Services halts work productivity sometimes entirely.

Knorr, E. (2022, July 22). *What is cloud computing? everything you need to know now*. InfoWorld. Retrieved November 8, 2022, from https://www.infoworld.com/article/2683784/what-is-cloud-computing.html

Knorr defines cloud computing as a technology that has become the "ideal way to deliver enterprise applications and launch new innovations." Knorr breaks down the components of cloud computing in how it works, what it does, and how we can use it. Highlighted, are the benefits of cloud computing and how its uses bleed into machine learning and the internet of things (IoT) connectivity.

Loeb, H. (2020, May 21). *Preventing crashes with automatic emergency braking*. Center for Injury Research and Prevention. (2021, April 1). Retrieved November 8, 2022, from https://injury.research.chop.edu/blog/posts/preventing-crashes-automatic-emergency-braking

Loeb's research in automatic emergency braking (AEB) defines this new technology as a growing success. AEB is shown to have promising results and has already been proven effective. AEB has prevented a large number of rear-end crashes in age ranges ranging from teens, to young adults, and older adults. With this major success, Loeb, a research scientist in robotics encourages parents to give teens access to vehicles with AEB technology.

Markets and Markets. (n.d.). *Autonomous / self-driving cars market size, share, trends and forecast by 2030*. Retrieved November 8, 2022, from https://www.marketsandmarkets.com/Market-Reports/near-autonomous-passenger-car-market-1220.html

Analyzing key concepts in market dynamics, Markets and Markets forecasts the autonomous car market to grow from 20.3 million units in 2021 to 62.4 million units by 2030. They highlight the increase in demand is garnered by the prominent safety features present in most modern cars today. The increase in market share is driven by the increase in

demand for safer, more efficient, and more convenient driving experiences. Markets and Markets also acknowledge the high system and component cost for the production of many more autonomous vehicles.

Microsoft. (2022, June 15). *Microsoft Services Agreement.* Retrieved November 8, 2022, from https://www.microsoft.com/en-us/servicesagreement

Microsoft's services agreement is the agreement between the user (you) and Microsoft. It defines and governs your use of Microsoft products and services.

Microsoft. (n.d.). *What is cloud computing? A beginner's guide: Microsoft azure*. What Is Cloud Computing? A Beginner's Guide | Microsoft Azure. Retrieved November 8, 2022, from https://azure.microsoft.com/en-us/resources/cloud-computing-dictionary/what-is-cloud-computing/#cloud-deployment-types

Microsoft formally and scientifically defines cloud computing and its intricacies. They define cloud computing as "the delivery of computing services — including servers, storage, databases, networking, software, analytics, and intelligence—over the Internet." Microsoft highlights the use cases of cloud computing in ways such as cloud-native applications, remote testing, and building applications. Four broad categories of cloud computing services known as the cloud computing "stack" are defined as: infrastructure as a service (IaaS), platform as a service (PaaS), serverless, and software as a service (SaaS).

The Mobile Economy. (2022, October 25). Retrieved November 8, 2022, from https://www.gsma.com/mobileeconomy/

The Mobile Economy provides insight into the smartphone market share and forecasts the increasing adoption of 5G technology in various countries. Asia and the United States lead this race and it is expected that 2 billion 5G connections will be established by 2025. Regional reports detail how mobile technologies contribute to the economy and the GDP of countries. It is strikingly shown that 53% of the population rely on and are connected through mobile devices.